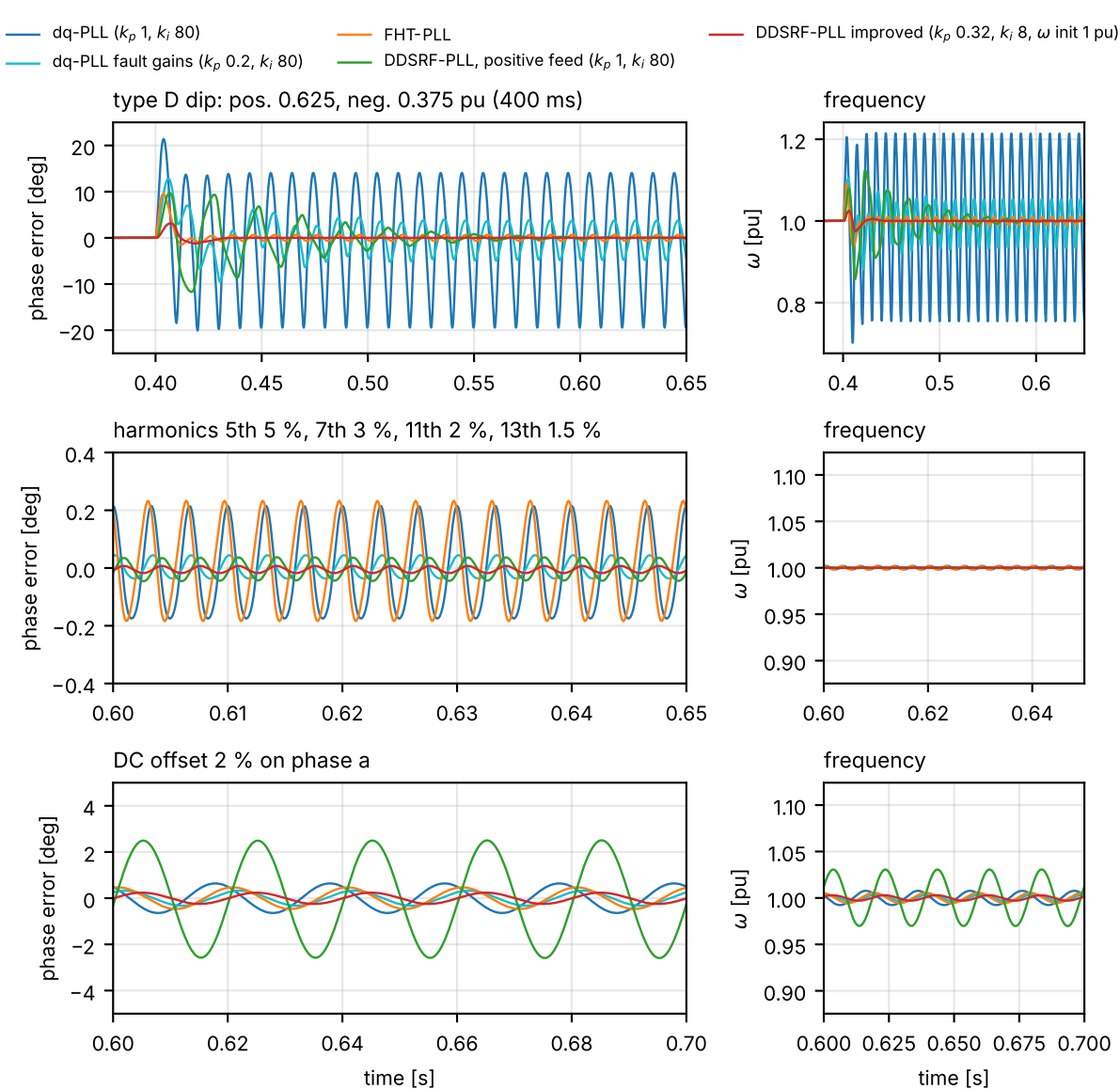

dq-PLL (k_p 1, k_i 80)
dq-PLL fault gains (k_p 0.2, k_i 80)
FHT-PLL
DDSRF-PLL, positive feed (k_p 1, k_i 80)
DDSRF-PLL improved (k_p 0.32, k_i 8, ω init 1 pu)
type D dip: pos. 0.625, neg. 0.375 pu (400 ms)
phase error [deg]
frequency
ω [pu]
harmonics 5th 5 %, 7th 3 %, 11th 2 %, 13th 1.5 %
DC offset 2 % on phase a
time [s]